

Markscheme

Mathematics: analysis and approaches
Practice question bank

173. (a)

Consider the system $x + 2y + z = 8$, $2x - y + 3z = 9$, $3x + y - 2z = -1$.

Solve the system for x , y and z .

eliminate x between pairs of equations, e.g. $(2) - 2 \times (1)$: $-5y + z = -7$ (M1) A1

and $(3) - 3 \times (1)$: $-5y - 5z = -25 \Rightarrow y + z = 5$ (M1) A1

solving these two equations simultaneously gives $y = 2$, $z = 3$ (M1)

back-substitute into (1): $x = 1$ A1

[6 marks]